PIPELINE IQ

DEVILS DEN

Team Members:

- Tanya Rishikesh
- Naman Mani
- Indraneel Chatterjee

Team Leader : naman.mani23@st.niituniversity.in
College Name: NIIT University

Agentic CI/CD Repair & Intelligent Release Orchestration

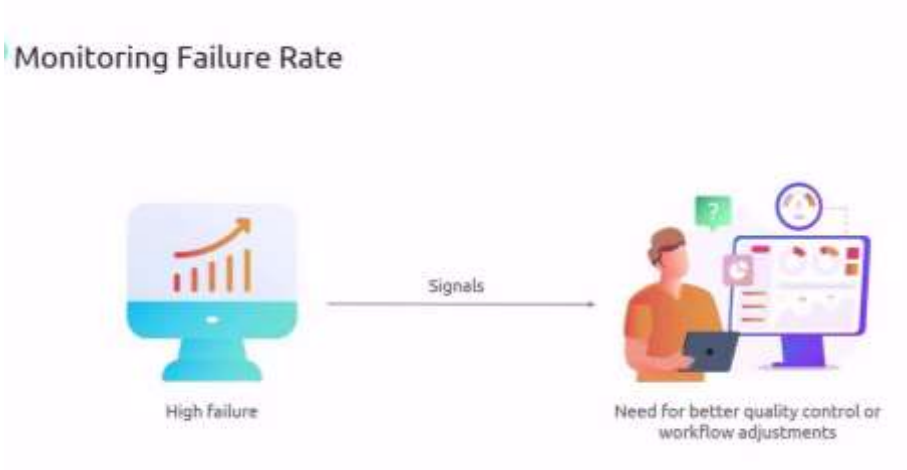
T1PS1

Modern software teams rely on complex CI/CD pipelines to deploy code frequently, but diagnosing pipeline failures often requires engineers to switch between multiple tools, such as version control, logs, test reports, and monitoring dashboards. As development velocity increases and AI-generated code becomes more common, the number of pipeline failures rises, leading to greater debugging effort and growing technical debt. At the same time, automation in DevOps pipelines can introduce security and reliability risks if autonomous systems operate with elevated permissions. The challenge is to build an Agentic CI/CD Orchestration System where autonomous agents monitor pipeline health, diagnose failures using contextual signals from commits, logs, and tests, automatically fix common issues, and enforce a human-in-the-loop governance layer for high-risk deployment decisions.

Why this matters?

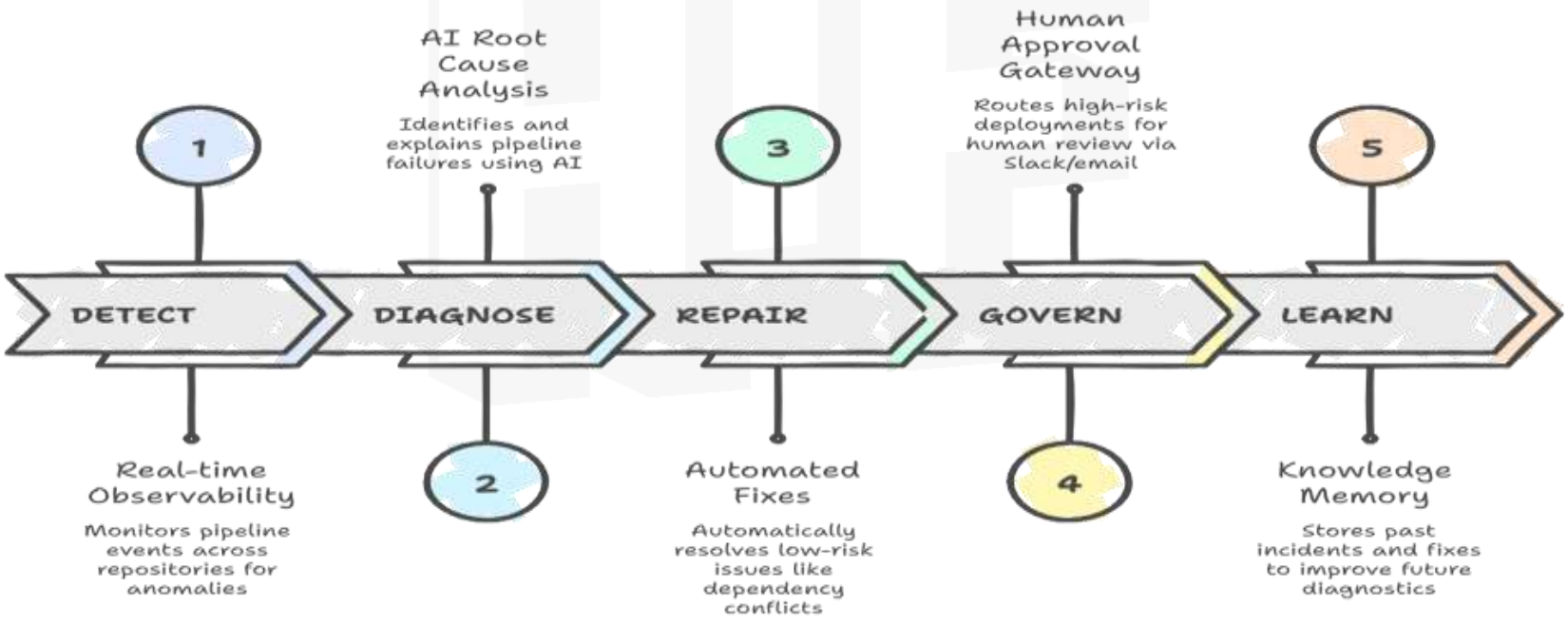
Growing Impact of the Problem

- Frequent deployments increase failure probability
- Debugging delays release cycles
- Engineering time shifts from building → fixing
- Automation without control can introduce risks



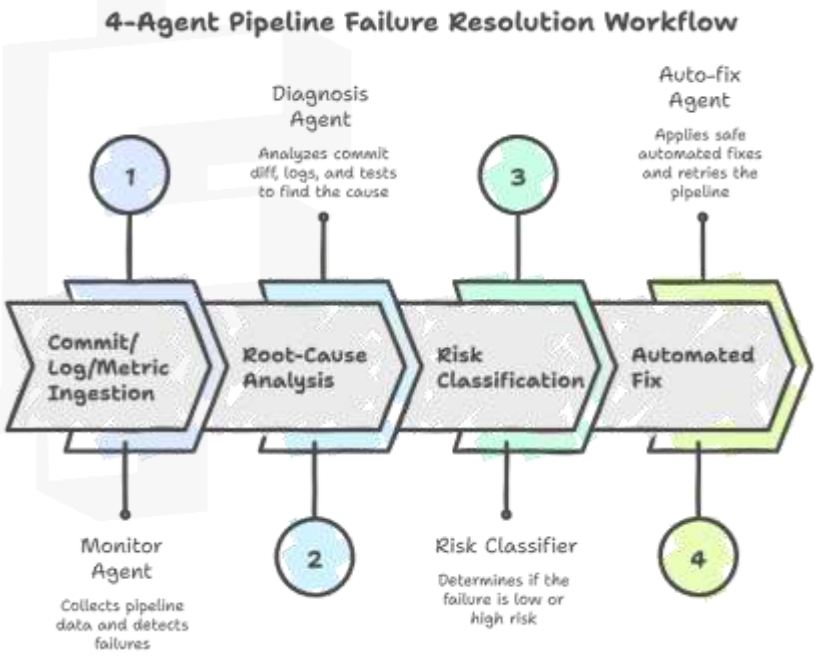
Solution Overview: **Pipeline IQ - Agentic CI/CD Manager**

A multi-agent orchestration system that monitors CI/CD pipelines in real-time, diagnoses failures using contextual signals from commits, logs, and tests then repairs common issues while enforcing human-on-the-loop governance for high-risk deployments actions. Built on event-driven architecture powered by Apache Kafka, independent AI agent process pipeline events asynchronously, enabling scalable monitoring, automated remedies and complete auditing of the fixes

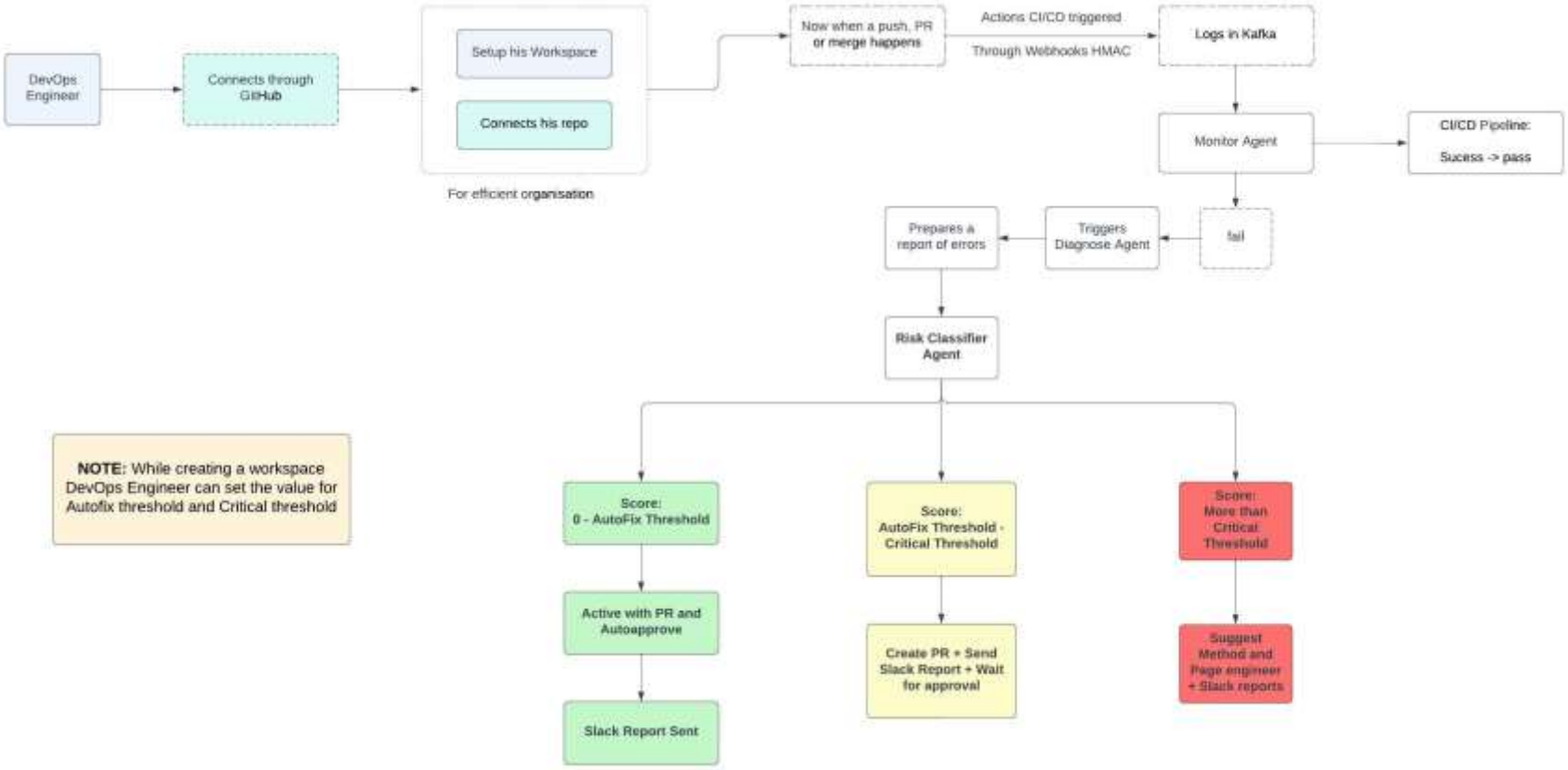


How PipelineIQ Agents Works?

- Pipeline IQ processes pipeline events in real time.
  It collects contextual data such as logs, test results, and commits, and uses AI agents to identify the root cause of failures.
- Based on the analysis, it either suggests fixes or applies them automatically, while ensuring human approval for critical actions.



Workflow


DevOps Engineer
Connects through GitHub
Setup his Workspace
Connects his repo
For efficient organisation
Now when a push, PR or merge happens
Actions CI/CD triggered
Through Webhooks HMAC
Logs in Kafka
Monitor Agent
CI/CD Pipeline:
Sucess -> pass
fail
Triggers Diagnose Agent
Prepares a report of errors
Risk Classifier Agent
Score: 0 - AutoFix Threshold
Score: AutoFix Threshold - Critical Threshold
Score: More than Critical Threshold
Active with PR and Autoapprove
Slack Report Sent
Create PR + Send Slack Report + Wait for approval
Suggest Method and Page engineer + Slack reports
NOTE: While creating a workspace DevOps Engineer can set the value for Autofix threshold and Critical threshold

Technical Stack

Frontend

- React 19
- Tailwind CSS
- WebSocket Client
- Recharts

Agentic Server

- Python 3.10-3.11
- FastAPI
- OpenAI GPT/Groq

Event & Messaging

- Apache Kafka
- Kafka-python
- Topic based communication agents

Pipeline Integration

- Github webhooks
- CI Simulator
- Log Stream Parser
- YAML Engine

Infra & Storage

- Docker
- SQLite
- Kafka Replay Store
- Fail Memory (MongoDB)

Governance & Notification

- Slack/Email
- Signed URL for approval
- Self APIs

Agent-wise Model Selection

Agent Name	Model Used
1.Monitor Agent	GPT 4o mini - Llama 3.3.7 (Backup model)
2. Diagnosis Agent	GPT OSS 120b - GPT 4.1 (Backup)
3. Risk Classification Agent	GPT 4o mini - Llama 3.3.7 (Backup model)
4. Auto Flx Agent	GPT OSS 120b - GPT 4.1 (Backup)

Safety & Governance in Pipeline IQ

- Pipeline IQ ensures that automation is applied responsibly by combining intelligent decision-making with human oversight.
- **Risk-Based Decision Making**
  Actions are classified based on environment, change impact, and failure severity
- **Human-in-the-Loop Approval**
  High-risk fixes require explicit approval before execution
- **Controlled Auto-Fix Mechanism**
  Limits repeated attempts to prevent infinite or unsafe actions
- **Audit & Transparency**
  All actions, decisions, and fixes are logged for traceability

How Pipeline IQ Stands Against Existing Tools

Capability	CodeRabbit	GitHub Copilot	PagerDuty	**Pipeline IQ**
Stage	Pre-merge	Code writing	Post-failure	CI/CD runtime
Understands pipeline context	✕	✕	✕	✓
Uses logs + tests + commits	✕	✕	✕	✓
Suggests fixes	✓	✓	✕	✓
Auto-fixes small issues	✕	✕	✕	✓
Human-in-loop governance	✕	✕	✕	✓
Resolves pipeline failures	✕	✕	✕	✓

Existing tools focus on code creation, review, or alerting.
Pipeline IQ focuses on **actively resolving CI/CD failures in real time** using contextual intelligence.

Applications & Expected Impact

Applications

Pipeline IQ is designed for environments where fast and reliable deployments are critical.

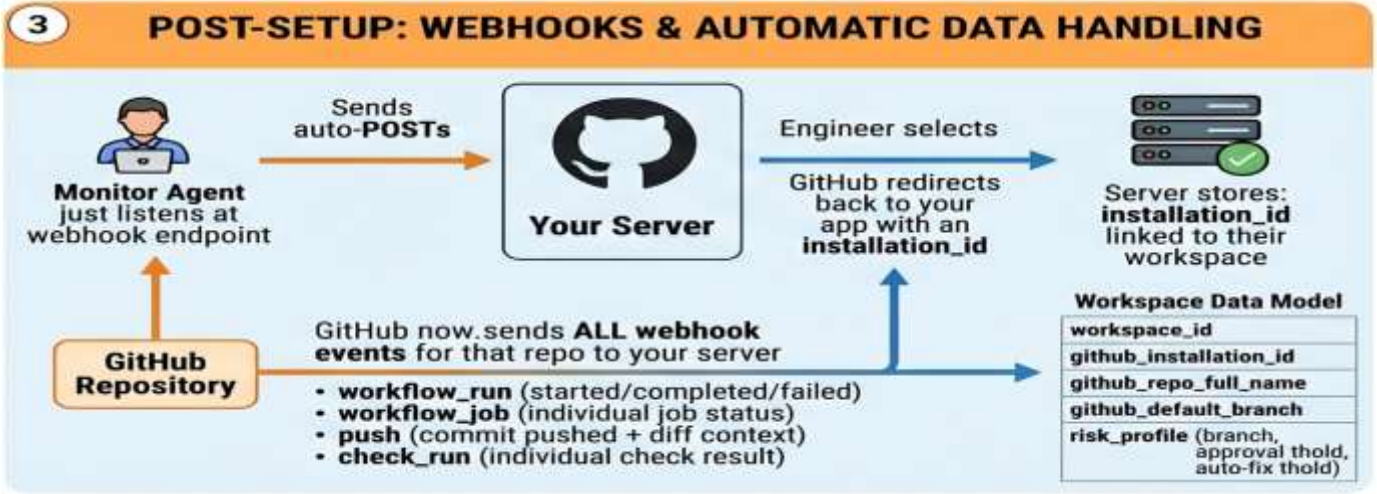
- **High-Frequency Deployment Teams**
  Quickly diagnose and resolve frequent pipeline failures
- **AI-Assisted Development Workflows**
  Handle unpredictable errors from AI-generated code
- **Scaling Startups**
  Reduce debugging effort with limited DevOps resources
- **Enterprise CI/CD Systems**
  Improve reliability across complex multi-service pipelines

Impacts

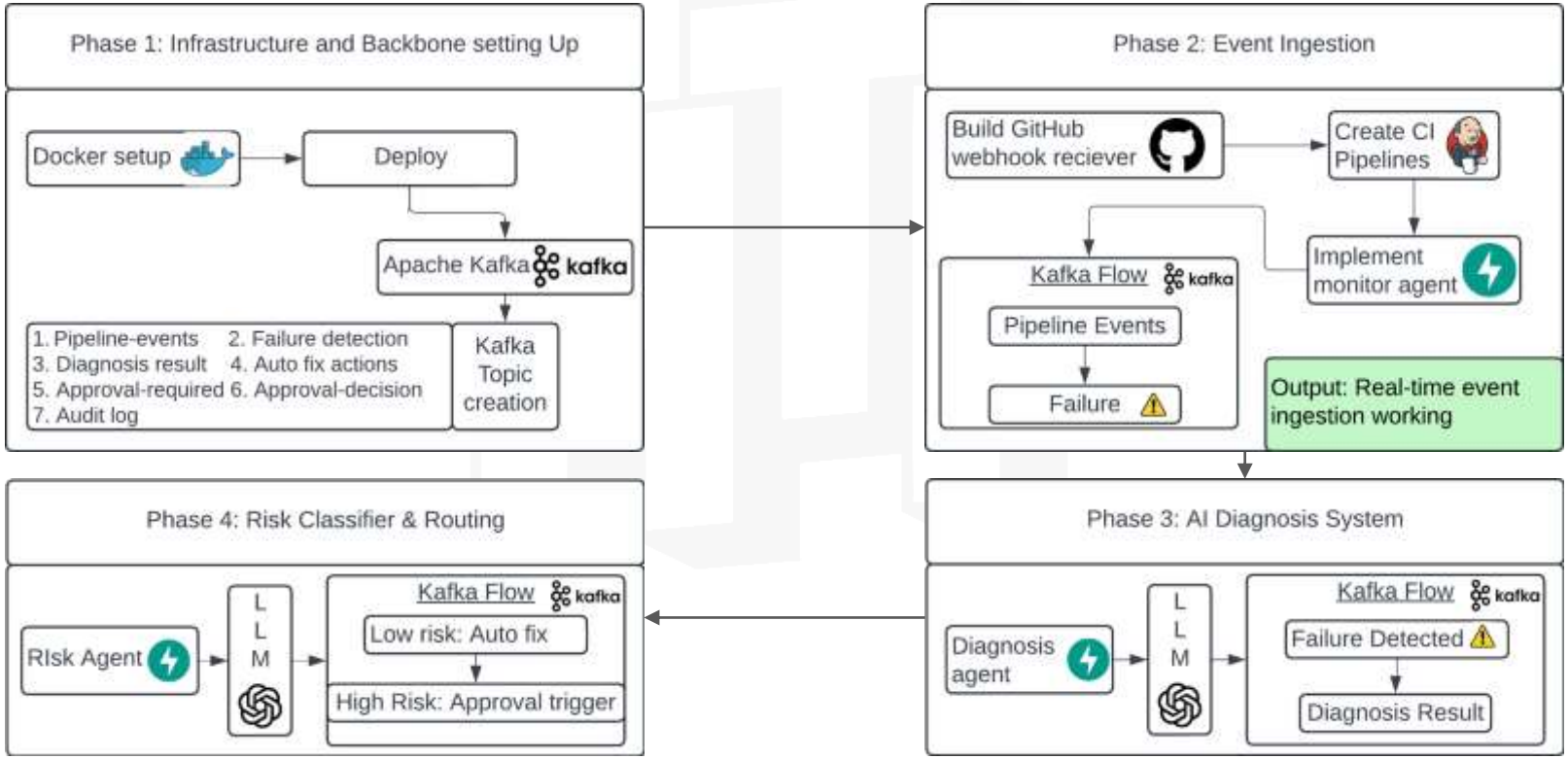
Pipeline IQ improves the efficiency and reliability of CI/CD pipelines by reducing manual intervention and accelerating failure resolution.

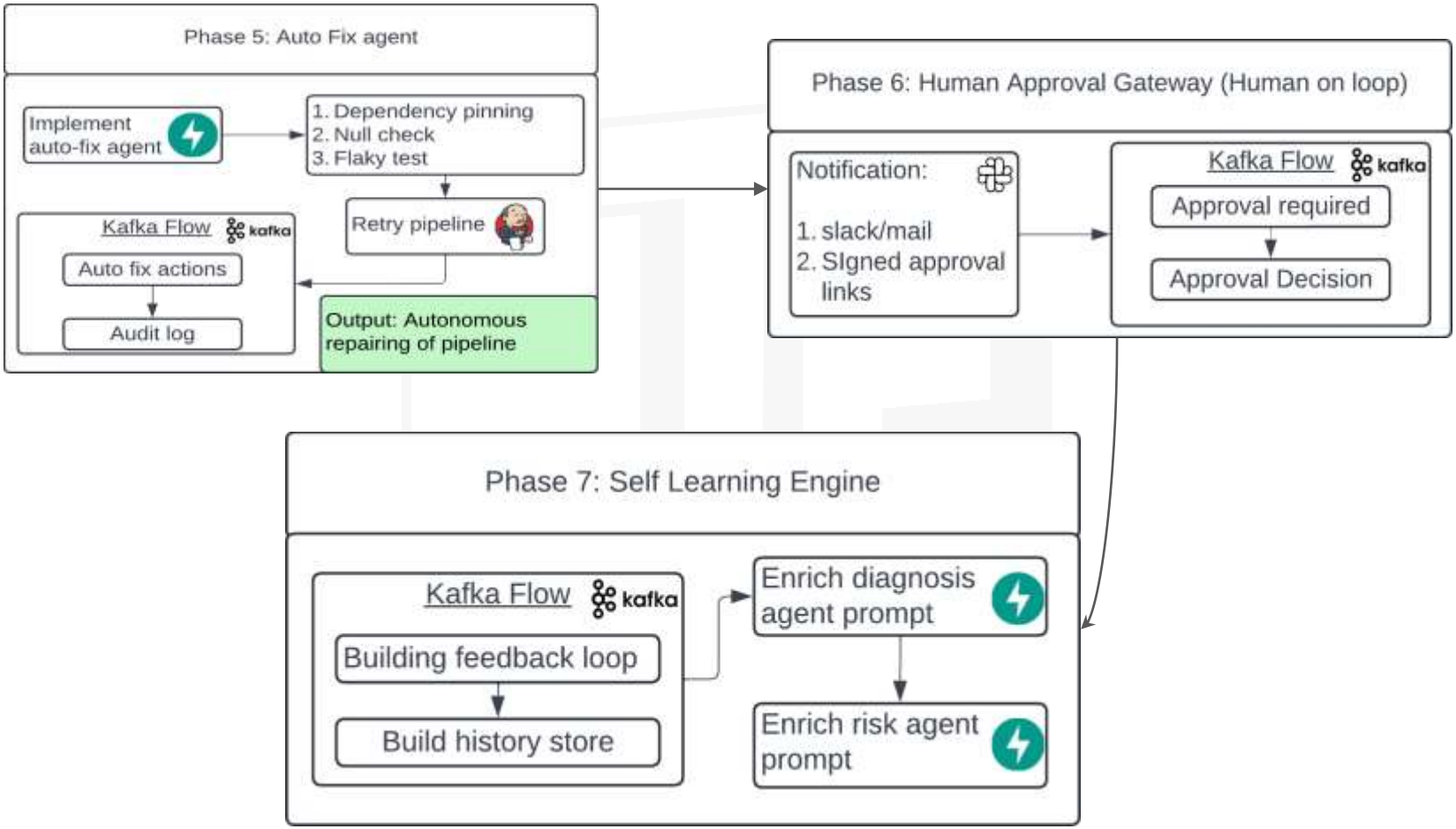
- **Faster Debugging**
  Reduces time spent identifying and fixing pipeline issues
- **Reduced Engineering Effort**
  Minimizes manual log analysis and repetitive fixes
- **Improved Reliability**
  Ensures more stable and consistent deployments
- **Safer Automation**
  Combines auto-fix with controlled human oversight

Deployment Strategy: User Onboarding



Deployment Strategy: Agent & Architecture Setup




Phase 5: Auto Fix agent
Implement auto-fix agent
1. Dependency pinning
2. Null check
3. Flaky test
Retry pipeline
Kafka Flow
kafka
Auto fix actions
Audit log
Output: Autonomous repairing of pipeline
Phase 6: Human Approval Gateway (Human on loop)
Notification:
1. slack/mail
2. SIgned approval links
Kafka Flow
kafka
Approval required
Approval Decision
Phase 7: Self Learning Engine
Kafka Flow
kafka
Building feedback loop
Build history store
Enrich diagnosis agent prompt
Enrich risk agent prompt

Key Implementations Strategy and Notes

- To avoid infinite loops, for the autofix, we tend to report the error to user if after a autofix, the same error repeats, or tend to form an level. Keeping the maximum threshold loop to be 2 in case of different errors every time.

- Risk classification metrics including following parameters:
  - Environment (Local/Dev server, Pre-production, Production)
  - File Type changed.
  - Change surface size.
  - Failure rate of service.
  - Test SIgnals.